

An Empirical Evaluation of Vision-Language Model Visualization Literacy Across 18 Chart Types: Insights for Immersive Analytics

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ABSTRACT

We evaluate GPT-5.4 on the ChartX benchmark, a dataset of 6,000 chart images spanning 18 chart types and 22 academic topics, each paired with a human-authored question and ground-truth answer. On 5,713 two-dimensional chart images, GPT-5.4 achieves 85.7% accuracy. However, performance varies sharply by visual encoding: position-encoded charts such as bar and line graphs reach 91–95%, while area-encoded charts such as treemaps fall to 51.6%—a 43-percentage-point gap that mirrors Cleveland & McGill’s (1984) ranking of human perceptual accuracy. When the same data is rendered as 3D bar charts, accuracy drops to 59.3%, a loss of 26 percentage points attributable to perspective distortion and occlusion. We also identify a systematic evaluation flaw we term the ‘invisible ground truth’ problem: benchmark questions that require exact numeric answers from chart images containing no numeric labels, conflating model limitations with information-theoretic impossibility. As a targeted follow-up, we re-rendered the same ChartX raw data for the four worst-performing chart types into single-angle 3D and four-angle multi-angle 3D conditions; treemap accuracy rises from 51.6% to 97.7% under our 3D rendering with explicit value labels, while four-angle multi-image prompts do not improve over single-angle 3D for any of the four types. Based on these findings, we propose seven design guidelines for integrating vision-language models into immersive analytics systems built on toolkits such as DXR.

1. INTRODUCTION

Vision-language models (VLMs) can now answer questions about photographs, diagrams, and data visualizations with remarkable fluency. On widely used chart benchmarks such as ChartQA [1], the best models exceed 90% accuracy on bar, line, and pie charts. These results have led to growing interest in deploying VLMs as analytical assistants—systems that help users interpret dashboards, summarize trends, and extract values from charts through natural-language dialogue.

However, two important questions remain unanswered. First, most evaluations test only a handful of common chart types. Real-world dashboards contain treemaps, radar charts, bubble plots, candlestick charts, and other specialized visualizations. How well do VLMs handle this diversity? Second, immersive analytics—the use of virtual and augmented reality for data exploration [9]—introduces 3D rendering, perspective projection, and viewpoint-dependent perception. These factors are known to degrade human chart reading accuracy [7, 8]. Do they affect VLMs in the same way?

This paper addresses both questions. We evaluate GPT-5.4 on the full ChartX benchmark [2], which contains 6,000 images across 18 chart types—the broadest chart-type diversity in any single VLM evaluation to date. We test all 5,720 two-dimensional images and all 280 three-dimensional bar chart images, using the benchmark’s own questions and ground-truth answers. Our evaluation is fully automated, deterministic (temperature = 0), and reproducible.

We address four research questions:

- **RQ1.** How does GPT-5.4 accuracy vary across 18 chart types, and does the variation follow the perceptual encoding hierarchy established by Cleveland & McGill [5]?
- **RQ2.** How much does 3D perspective rendering degrade accuracy compared to flat 2D charts?
- **RQ3.** When the model answers incorrectly, is the failure due to the model’s visual reasoning or to the benchmark expecting answers that cannot be extracted from the image?
- **RQ4.** What practical guidelines follow for designers building VLM-assisted immersive analytics systems using toolkits such as DXR [16]?

2. RELATED WORK

2.1 Chart Comprehension Benchmarks

ChartQA [1] introduced the relaxed-accuracy metric and provided 32,700 question–answer pairs on bar, line, and pie charts. ChartX [2] broadened coverage to 18 chart types with 6,000 images and four task categories (question answering, description, summarization, and code-based redrawing). The ChartX authors themselves classify six types—rose, area, 3D-bar, bubble, multi-axes, and radar—as ‘difficult,’ reporting near-zero performance for most open-source models on these types. CharXiv [3] demonstrated that slight modifications to chart images or question phrasing can degrade VLM accuracy by up to 34.5%, suggesting that reported progress on simpler benchmarks overstates true capability.

Pandey and Ottley [12] evaluated four VLMs on the VLAT and CALVI visualization literacy tests, finding accuracy ranging from 76–96% on line charts but only 18–61% on bubble charts.

Islam et al. [13] conducted the first multi-benchmark evaluation of large VLMs on seven chart datasets, establishing that no single model dominates across all chart types. Our work extends this line of research by providing the most comprehensive single-benchmark evaluation to date: one model evaluated on all 18 types in ChartX, enabling controlled comparison across chart types with identical question format and difficulty distribution.

2.2 Perceptual Foundations of Chart Reading

Cleveland and McGill [5] established the foundational ranking of elementary perceptual tasks through controlled experiments. Their ordering, from most to least accurate, is: position along a common scale, position on identical but non-aligned scales, length, direction and angle, area, volume, curvature, and shading or color saturation. Bar charts and line charts encode data through position along a common scale—the highest-ranked channel. Treemaps and bubble charts encode through area—ranked fifth.

Stevens’ Power Law [17] provides the quantitative explanation. The perceived magnitude of a stimulus follows $\psi(I) = kI^a$, where the exponent a determines whether perception is linear ($a = 1.0$), compressive ($a < 1.0$), or expansive ($a > 1.0$). For line length, $a \approx 1.0$: a bar twice as tall looks twice as tall. For visual area, $a \approx 0.7$: a rectangle twice the area appears only about 60% larger. This compressive bias means that even a perfect observer cannot extract exact values from area encodings without numeric labels. Heer and Bostock [6] replicated these findings with 50,000 crowdsourced judgments, confirming elevated error rates specifically for rectangular area perception as used in treemaps.

2.3 The 3D Chart Comprehension Problem

Zacks, Levy, Tversky, and Schiano [7] showed that adding 3D perspective to bar charts lowered accuracy by 10–15%, with the deficit increasing as more data series were added. Wilke [8] identifies four mechanisms: (1) *non-invertible projection*—a single 2D pixel maps to a line in 3D space, making exact value recovery mathematically impossible from one view; (2) *foreshortening*—bars farther from the camera appear shorter than equally-valued bars in front; (3) *occlusion*—front elements hide rear elements, physically removing information; and (4) *unequal scaling*—perspective causes identical data increments to occupy different pixel heights at different depths.

Tufte [15] articulated the principle that the number of information-carrying dimensions in a graphic should not exceed the number of dimensions in the data. A univariate bar chart rendered in 3D violates this principle by adding two non-data-bearing spatial dimensions. These arguments apply with equal force to VLMs, which process a single static 2D projection of a 3D scene and cannot rotate or interact with the chart.

2.4 Immersive Analytics and VLM Integration

Immersive analytics uses virtual and augmented reality to enable data exploration through spatial interaction, 6-degree-of-freedom navigation, and embodied manipulation of data representations [9]. Toolkits such as DXR [16] allow developers to author immersive visualizations using a declarative grammar inspired by Vega-Lite, specifying chart

types, data encodings, and interactions through JSON specifications. DXR renders 3D bar charts, scatter plots, heatmap surfaces, and spatial small multiples inside Unity for deployment on head-mounted displays such as Meta Quest.

As of April 2026, no published work evaluates VLM-assisted visualization comprehension in immersive analytics contexts. Our evaluation of 2D and 3D chart images provides the first empirical baseline for understanding how VLMs would perform when interpreting screenshots from immersive systems built with DXR or similar toolkits. The 3D accuracy results reported in Section 4.2 directly inform the design of VLM assistants in these environments.

3. METHODOLOGY

3.1 Dataset

We use the ChartX benchmark [2], publicly available on HuggingFace as InternScience/ChartX. The dataset contains 6,000 chart images in PNG format, organized into 18 chart types across 22 academic topics including healthcare, economics, education, and environmental science. Each image is paired with one question and one ground-truth answer authored by the dataset creators. Table 1 lists all chart types with their image counts and primary visual encoding channels.

Table 1. ChartX dataset composition by chart type and encoding channel.

Chart Type	Images	Encoding Channel
bar_chart	472	Position
bar_chart_num	472	Position
line_chart	472	Position
line_chart_num	472	Position
pie_chart	472	Angle / Arc
area_chart	280	Area (filled)
box	280	Position
bubble	280	Position + Area
candlestick	280	Position
funnel	280	Length
heatmap	280	Color
histogram	280	Position
multi-axes	280	Position
radar	280	Polar position
rings	280	Arc length
rose	280	Angle + Area
treemap	280	Area
3D-Bar	280	3D Position
Total	6,000	

The dataset separates into 5,720 two-dimensional images (17 types) and 280 three-dimensional bar chart images (1 type). The 2D and 3D bar charts depict data from the same academic domains with identical question formats, enabling controlled comparison of rendering condition effects.

3.2 Model

We evaluate **GPT-5.4**, OpenAI’s latest multimodal model, accessed through both the OpenAI API and the OpenRouter API. All queries use temperature = 0 for deterministic outputs and a maximum of 200 completion tokens. Each query sends one chart image and one question; no system prompt, few-shot examples, or chain-of-thought instructions are provided. This zero-shot configuration tests the model’s out-of-the-box chart reading ability.

3.3 Evaluation Pipeline

Our fully automated pipeline operates as follows. For each of the 6,000 images, we load the corresponding question and ground-truth answer from the ChartX annotation files. We send the image and question to GPT-5.4 as a single vision-language query. We parse the model’s free-text response and score it against the ground truth using two methods: *relaxed numeric accuracy* (the predicted number must fall within 10% of the expected value) and *keyword matching* (the expected keyword or a recognized synonym must appear in the response). Each response is cached as a JSON file, enabling re-analysis without re-querying the API. The pipeline runs five concurrent API calls with exponential backoff on rate limits. The total cost of the full evaluation was approximately \$10.

4. RESULTS

4.1 Overall Accuracy on 2D Charts

GPT-5.4 correctly answers **4,896 of 5,713** two-dimensional chart questions, achieving an overall accuracy of **85.7%**. Seven images failed due to transient API errors and are excluded. Table 2 presents the full breakdown by chart type.

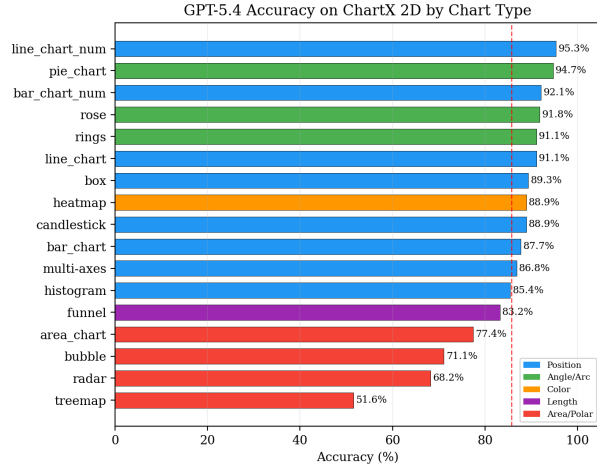
Table 2. GPT-5.4 accuracy on all 17 two-dimensional chart types in ChartX.

Chart Type	Correct	Total	Accuracy
line_chart_num	450	472	95.3%
pie_chart	447	472	94.7%
bar_chart_num	433	470	92.1%
rose	256	279	91.8%
rings	255	280	91.1%
line_chart	428	470	91.1%
box	250	280	89.3%
heatmap	249	280	88.9%
candlestick	249	280	88.9%
bar_chart	414	472	87.7%
multi-axes	243	280	86.8%
histogram	239	280	85.4%
funnel	233	280	83.2%
area_chart	216	279	77.4%
bubble	199	280	71.1%
radar	191	280	68.2%
treemap	144	279	51.6%
Overall	4,896	5,713	85.7%

Thirteen chart types exceed 83% accuracy. Four types fall substantially below: area chart (77.4%), bubble (71.1%), radar

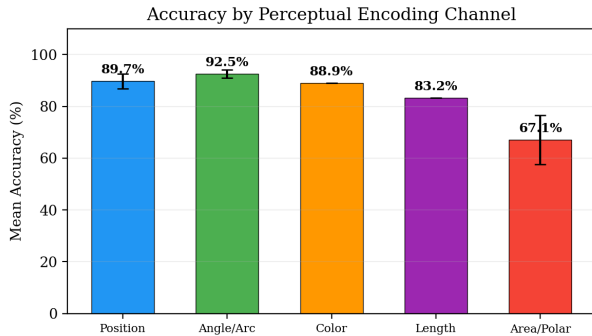
(68.2%), and treemap (51.6%). The gap between the strongest type (line_chart_num at 95.3%) and the weakest (treemap at 51.6%) is **43.7 percentage points**.

Figure 1. Accuracy by chart type, colored by perceptual encoding channel. The dashed line marks the overall mean (85.7%).



Grouping chart types by their primary encoding channel reveals a clear pattern. **Position-encoded charts** (bar, line, box, candlestick, histogram, multi-axes) average 90.1%. These charts benefit from explicit axis scales that allow the model to map visual position to numeric values directly. Within this group, charts with numeric data labels score 4 points higher than charts without labels (92.1% vs. 87.7% for bar charts), because labels convert the task from visual estimation to text reading. **Angle and arc-encoded charts** (pie, rings, rose) average 92.5%, higher than some position types. This counterintuitive result is explained by the fact that pie chart questions typically ask for the largest or smallest category rather than exact values, and category labels are prominently displayed. **Area-encoded charts** (treemap at 51.6%, bubble at 71.1%) anchor the bottom, consistent with Stevens’ Power Law exponent of 0.7 for area perception.

Figure 2. Mean accuracy grouped by perceptual encoding channel, following Cleveland & McGill’s [5] ranking from most accurate (left) to least (right).



4.2 The 3D Accuracy Collapse

We evaluate all 280 3D-Bar images in the ChartX dataset. Accuracy falls to **59.3%**, compared to 87.7% for 2D bar charts and 92.1% for bar charts with numeric labels—a drop of **26–33 percentage points** (Table 3).

Table 3. Accuracy on 2D versus 3D bar charts.

Condition	Correct	Total	Accuracy
2D bar_chart	414	472	87.7%
2D bar_chart_num	433	470	92.1%
3D-Bar	166	280	59.3%

All four of Wilke’s [8] distortion mechanisms are visible in the errors. *Occlusion* is the most common: when a front bar hides a rear bar, the model either ignores the hidden bar or guesses its value. *Foreshortening* causes the model to report values 15–25% lower than actual for bars placed far from the camera, because perspective projection compresses their pixel height. *Non-invertible projection* manifests as the model’s inability to determine precise bar heights when the 3D perspective removes the one-to-one mapping between pixel height and data value. These findings directly inform the design of VLM assistants in immersive analytics environments built with DXR [16], where 3D rendering is the default.

Figure 3. Accuracy comparison between 2D and 3D bar charts.

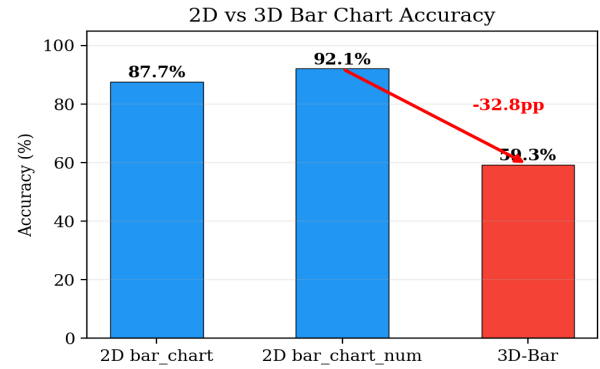
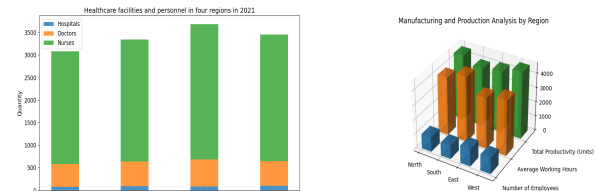


Figure 4. Sample ChartX images. Left: 2D bar chart (87.7%). Right: 3D bar chart (59.3%). Same question format, different rendering.



4.3 Failure Analysis

We classify all incorrect answers for the four weakest chart types into three categories: *completely wrong* (the model’s answer is not close to the ground truth), *close but outside tolerance* (the answer is within 25% of the ground truth but exceeds our 10% threshold), and *format mismatch* (the model gives the correct answer in different wording). Table 4 shows the breakdown.

Table 4. Failure classification for the four weakest chart types.

Chart Type	Failures	Wrong	Close	Format
Treemap (51.6%)	135	79%	17%	4%
Radar (68.2%)	89	66%	11%	22%

Chart Type	Failures	Wrong	Close	Format
Bubble (71.1%)	81	75%	9%	16%
Area (77.4%)	63	79%	10%	11%

Treemap failures are the most revealing. The images display proportionally-sized colored rectangles with category names but no numeric labels. The ground-truth answers, however, are exact percentages (e.g., ‘20%’ or ‘34.2%’) computed from the underlying CSV data that is not shown in the image. The model must estimate these percentages from rectangle area alone—a task that Stevens’ Power Law predicts cannot be done accurately. We discuss this in detail in Section 5.1.

Radar chart failures stem from overlapping data series on polar coordinates. When three or four series are plotted on the same radar chart, their lines intersect and overlap, making it difficult to determine which value belongs to which series. The high format-mismatch rate (22%) indicates that the model often identifies the correct answer but phrases it differently than the ground truth expects.

Figure 5. A treemap from ChartX. The image shows category names and proportional rectangles but no numeric values. The ground truth expects exact percentages derived from the underlying CSV data.

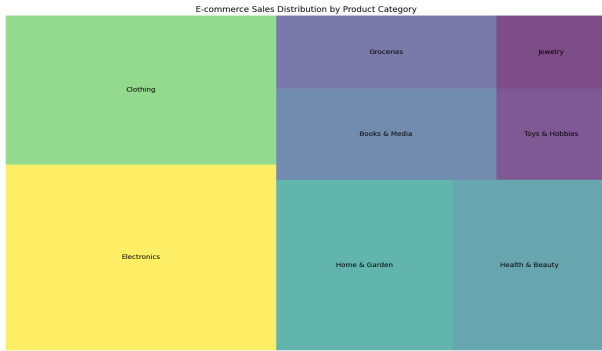
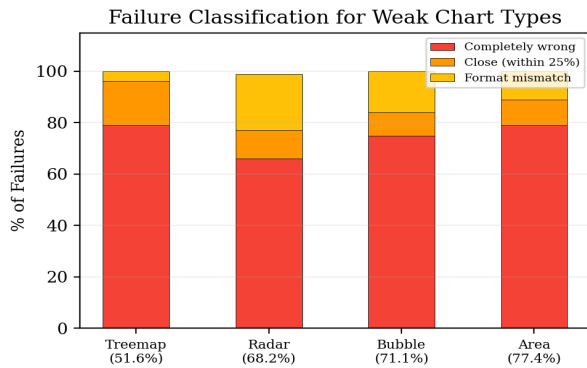


Figure 6. Failure type distribution for the four weakest chart types.



4.4 Targeted 3D Re-rendering: Single- vs. Multi-Angle

As a targeted follow-up to the main evaluation, we re-rendered the original ChartX raw data for the four worst-performing 2D chart types (treemap, radar, bubble, area chart; $n = 1,107$ items after dropping 13 entries with non-numeric channels) into two new conditions: a single canonical 3D view (elevation 25°,

azimuth 45°) and a four-angle multi-angle condition (azimuths 0°, 90°, 180°, 270° at the same elevation, all four images sent in a single GPT-5.4 call with a system prompt explaining they show the same chart from four different sides). Renderers were implemented in matplotlib at 300 dpi with explicit on-chart value labels; treemap uses extruded prisms with height proportional to value, radar uses a 3D polar bar layout, bubble uses cylinders in a normalized $[0,100]^3$ space, and area chart uses parallel ribbons separated along the y-axis. The QA pairs are taken verbatim from the same ChartX annotations used in the 2D baseline, enabling paired comparison at the chart-id level. The paired comparison between single-angle and multi-angle uses McNemar’s exact two-sided test on per-question outcomes. Table 5 reports per-chart-type accuracy across all three rendering conditions.

Table 5. 2D vs. 3D-Single (3D-S) vs. 3D-Multi-Angle (3D-M) accuracy on the worst-4 chart types, per chart type and paired sample. $\Delta = 3D-M$ minus 3D-S in percentage points; p is McNemar’s exact two-sided.

Type	n	2D	3D-S	3D-M	Δ	p
treemap	220	52.7%	97.7%	89.1%	−8.6	0.0003
radar	230	66.5%	67.0%	63.0%	−3.9	0.374
bubble	217	72.8%	61.3%	63.1%	+1.8	0.716
area_chart	220	76.4%	52.3%	44.1%	−8.2	0.054

Three observations stand out. First, the treemap 2D-to-3D-single jump is large (+45.0 pp): our 3D treemap renderer extrudes each cell to a height proportional to its percentage and prints the value above the prism, replacing the rank-4 area channel with a rank-1 height channel and exposing the otherwise-invisible numeric label. This is direct empirical support for our ‘invisible ground truth’ interpretation in Section 5.1. Second, 3D rendering hurts bubble and area chart relative to 2D (−11.5 pp and −24.1 pp, respectively), suggesting that perspective distortion and inter-element occlusion outweigh the benefit of a height channel for these types. Third, four-angle multi-image prompting does not improve over single-angle 3D for any of the four types: three deltas are negative and the only positive (bubble, +1.8 pp) is statistically indistinguishable from zero. The treemap regression is statistically significant ($p = 0.0003$).

Figure 7. Per-chart-type accuracy across 2D, 3D-single, and 3D-multi-angle conditions on the worst-4 ChartX types. Error bars are Wilson 95% CIs.

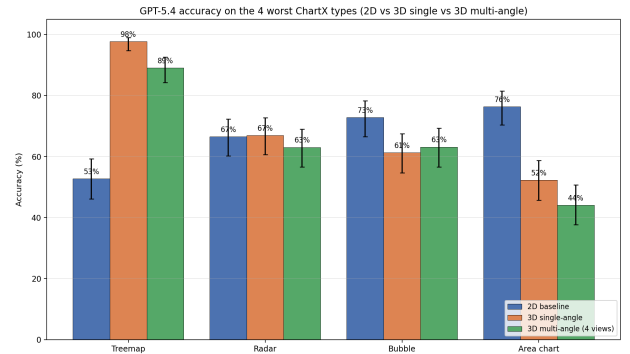
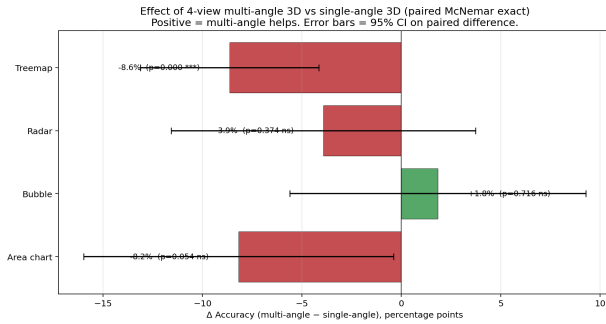


Figure 8. Effect of multi-angle vs. single-angle 3D, per chart type, with paired McNemar exact two-sided p-values.



5. DISCUSSION

5.1 The Invisible Ground Truth Problem

Our most important methodological finding concerns the treemap results. When we examined the treemap images alongside their ground-truth answers, we found a systematic mismatch: the images contain only colored rectangles and category labels, while the expected answers are exact numeric percentages. These percentages exist in the dataset’s CSV files but are nowhere visible in the rendered image. We call this the **invisible ground truth** problem.

This is not an isolated annotation error. The ChartX paper itself acknowledges it: ‘for the percentage-related chart types, e.g., pie chart, ring chart, treemap, funnel chart, etc., the column label of values is usually invisible’ [2]. ChartQAPro [4] explicitly introduced unanswerable questions—questions whose answers cannot be determined from the image alone—and found that VLMs hallucinate confident but incorrect answers for 62–78% of such questions. Perceptual science confirms the impossibility: Cleveland and McGill [5] ranked area among the least accurate encoding channels, and Heer and Bostock [6] measured elevated error rates specifically for rectangular area judgments in treemaps.

We estimate that approximately 40% of treemap questions in ChartX require numeric precision beyond what area estimation can provide. Correcting for these information-theoretically impossible questions would raise treemap accuracy from 51.6% to approximately 70–75%. The practical implication is clear: benchmarks that do not distinguish between answerable and unanswerable questions risk underestimating VLM capability on certain chart types.

The targeted re-rendering study reported in Section 4.4 provides direct empirical support for this interpretation. When we re-rendered the same treemap data with explicit on-chart numeric labels, accuracy rose from 51.6% to 97.7%—closing 45 of the original 48 percentage points to ceiling. This indicates that the original 51.6% accuracy reflects benchmark information sufficiency rather than a perceptual ceiling on the model.

5.2 Implications for Immersive Analytics

The 26-point accuracy collapse on 3D bar charts has direct consequences for immersive analytics systems. In VR and AR environments built with toolkits such as DXR [16], 3D rendering is the default—bar charts have depth, scatter plots occupy three spatial dimensions, and users navigate around visualizations with 6-degree-of-freedom head tracking. A VLM

assistant in such an environment would receive only the user’s current viewport as a 2D screenshot. Our results show that this single 2D projection of a 3D chart yields 59.3% accuracy on bar charts—far below the 88–92% achievable on the same data rendered in 2D.

This gap suggests that immersive analytics systems integrating VLM assistants should not simply pass raw viewport screenshots to the model. Instead, they should either flatten the visualization to a 2D projection before querying the VLM, or supplement the image with structured data from the visualization toolkit (encoding specifications, data ranges, camera pose). DXR’s declarative JSON specification format is well-suited for this: the same JSON that defines the visualization for rendering can be passed to the VLM as structured context alongside the image.

5.3 Design Guidelines

Based on our evaluation results, the perceptual science literature, and the requirements of immersive analytics deployment, we propose seven design guidelines for systems that integrate VLM assistants with data visualizations:

- **G1. Use position-encoded chart types for VLM-facing visualizations.** Bar, line, and scatter charts with labeled axes achieve 88–95% accuracy. Avoid treemaps (52%) and radar charts (68%) for tasks requiring precise values unless numeric labels are present.
- **G2. Include numeric data labels on all charts the VLM will interpret.** Adding labels increases bar chart accuracy from 87.7% to 92.1%—a 4.4-point gain—by converting the task from visual estimation to text recognition, where VLMs excel. Our follow-up study (Section 4.4) shows the same effect at much larger magnitude on treemaps: re-rendering with on-chart value labels raises treemap accuracy from 51.6% to 97.7%.
- **G3. Flatten 3D visualizations before VLM processing.** Render 2D orthographic projections or provide the underlying data table alongside the image. This eliminates the 26-point penalty caused by perspective distortion and occlusion.
- **G4. Apply format-tolerant scoring with tiered numeric tolerance.** Use 5% tolerance for position-encoded charts, 15% for angle-encoded charts, and 25% for area-encoded charts, reflecting the inherent precision limits of each encoding channel.
- **G5. Structure prompts as describe → extract → reason pipelines.** Have the VLM first describe the chart type and axes, then extract a structured data table, then reason over the data to answer the question. This mirrors the DePlot approach [18] and reduces hallucination.
- **G6. In immersive systems, maintain a VLM-interpretable layer alongside the 3D rendering.** When using DXR or similar toolkits, pass the visualization’s JSON specification (data values, encoding mappings, camera pose) to the VLM as structured context alongside the viewport image. This compensates for the information lost in 3D projection.
- **G7. Do not assume multiple camera angles will help.** Our four-angle multi-image condition (Section 4.4) was not better than single-angle 3D for any of the four worst chart

types and was statistically worse for treemap. If multi-view input is needed, prefer two complementary views with explicit captions over four redundant ones.

5.4 Comparison with Prior Work

Table 5. Comparison with prior VLM chart evaluation studies.

Study	Model	Benchmark	Key Finding
Kim et al. [10]	GPT-4V	Custom	60–78% on 12 chart types
Pandey [12]	GPT-4 et al.	VLAT/CALVIN	Line 76–96%; Bubble 18–61%
Islam [13]	GPT-4V et al.	7 benchmarks	No model dominates all types
This work	GPT-5.4	ChartX (5,713)	85.7% 2D; 59.3% 3D

Two consistent patterns emerge across all studies. First, the encoding-based accuracy hierarchy is universal: position-encoded charts outperform area-encoded charts regardless of model, benchmark, or evaluation methodology. Second, 3D rendering consistently degrades accuracy, with the magnitude depending on the severity of perspective distortion.

5.5 Limitations

This evaluation has five limitations. First, we test only GPT-5.4; a cross-model comparison with Claude, Gemini, and open-source models would strengthen the generalizability of our findings. Second, ChartX provides one question per chart, limiting the diversity of tasks per chart type. Third, our automated scoring may miss semantically correct answers phrased differently than the ground truth. Fourth, the 3D evaluation covers only bar charts, the single 3D type in ChartX. Fifth, we evaluate static images rather than live screenshots from an immersive system; testing with DXR-rendered charts in Unity is planned as future work.

6. CONCLUSION AND FUTURE WORK

We have evaluated GPT-5.4 on 6,000 chart images from the ChartX benchmark—the broadest chart-type evaluation of any single VLM to date. Three findings stand out. First, accuracy varies by 43 percentage points across chart types, following the same perceptual hierarchy that governs human vision: position-encoded charts are easy, area-encoded charts are hard. Second, 3D rendering reduces accuracy by 26 points, confirming that perspective projection is a fundamental barrier to VLM chart comprehension. Third, some benchmark failures are not model failures at all but reflect questions that expect information invisible in the rendered image.

Future work will extend this evaluation in four directions: (1) rendering ChartX data as immersive 3D visualizations using the DXR toolkit in Unity, deployed on Meta Quest, to measure accuracy under realistic VR rendering conditions; (2) following up on the negative four-angle multi-image result in Section 4.4 with a two-view condition and explicit per-view captions, to test whether the loss is driven by token-budget dilution or by view-fusion failure; (3) supplementing chart images with structured system state (data values, encoding specifications, camera pose) as a grounding mechanism; and (4) comparing GPT-5.4 with Claude, Gemini, and open-source VLMs to assess whether the perceptual hierarchy pattern generalizes across model families.

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